

QP CODE: 24800325



Reg No :

Name :

M.C.A. DEGREE EXAMINATION, JANUARY 2024

First Semester

Faculty of Technology & Applied Science

Master of Computer Application

CORE - MCACT102 - DIGITAL LOGIC & COMPUTER ORGANIZATION

2020 Admission Onwards

2DC309FC

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. What is BCD code ? Give examples.
2. Illustrate the use of error correcting codes.
3. What is EBCDIC code?
4. Convert the following Boolean expression into standard SOP form $AB'C + A'B' + ABC'D$
5. What is a flip-flop?
6. List out basic types of shift registers.
7. Discuss the different ways to arrange byte addresses across memory words.
8. Give the purpose of MFC signal.
9. Illustrate I/O interface for an input device.
10. Explain the function of Interrupt Service Routine.
11. Explain vector processors.
12. Differentiate between superscalar and super pipelined systems.

(10×3=30 marks)





Part B

Answer **all** questions

Each question carries **9** marks

13. a) Perform subtraction using 1's complement method on the binary and explain the steps.
- i) 10101 - 00110
 - ii) 00110 - 01001
 - iii) 00011 - 01100

OR

- b) Explain how to detect and correct errors in the following odd parity Hamming code 0101101.
14. a) Solve the following using K-Map. $F(A,B,C, D) = \sum(1,3,9,11,4, 5, 12, 13, 10,14)$.
- OR
- b) Explain the four types of shift registers.

15. a) Draw and explain single bus organization of datapath inside a processor.

OR

- b) Explain Booth Algorithm with an example.
16. a) Explain the various memory classifications.
- OR
- b) Write notes on a)Secondary memory b) ROM.
17. a) Explain in detail the following SIMD parallel structures: Array Processors, Vector Processors.

OR

- b) Explain the interconnection structures used in multiprocessors .

(5×9=45 marks)

